

1. PROBABILITY TOOLKIT

Conditioning & Bayes
P(A|B)=P(A∩ B)/P(B)
P(A∩ B)=P(A|B)P(B)
Total probability, partition {Bj}:
P(A)=∑i P(A|Bi)P(Bi)
Bayes (invert the conditioning):
P(Bk|A)=P(A|Bk)P(Bk)/∑i P(A|Bi)P(Bi)
Independence: P(A∩ B)=P(A)P(B).
Expectation & variance
E[aX+bY]=aE[X]+bE[Y]
Var(X)=E[X²]-(E[X])²
Var(aX+b)=a² Var(X)
Independent X,Y: E[XY]=E[X]E[Y] and
Var(X+Y)=Var(X)+Var(Y).
Cov(X,Y)=E[XY]-E[X]E[Y]
E[1A]=P(A)
Linearity needs no independence. Independent variables
have Cov=0 and E[XY]=E[X]E[Y]; zero covariance alone does
not imply independence.
Indicators
1A^c=1-1A, 1A∩c B=1A1B
Use indicators to count visits, successes, or event
occurrences; sum them and take expectations.
Common distributions
Bernoulli(p): mean p, var p(1-p)
Binomial(n,p): mean np, var np(1-p)
Geometric(p): P(X=k)=(1-p)^{k-1}p, k≥1; mean 1/p, var (1-p)/p²
Poisson(λ): P=e^{-λ}λ^k/k!; mean=var=λ

2. COND. EXPECTATION

E[X|Y]=g(Y) is a random variable (function of Y).
Tower / total expectation
E[X]=E[E[X|Y]]
E[X]=∑y E[X|Y=y]P(Y=y)
Law of total variance
Var(X)=E[Var(X|Y)]+Var(E[X|Y])
Useful identities
E[Xg(Y)|Y]=g(Y)E[X|Y]
Take known functions of Y outside the conditional.
3. RANDOM SUMS
N random count, Xi iid, independent of N. S_N=X₁+...+X_N.
E[S_N]=E[N]E[X]
Var(S_N)=E[N]Var(X)+Var(N)(E[X])²
Derive bv conditioning on N (tower + total variance).
4. GEN. FUNCTIONS (PGF)

For N-valued X:
φ(s)=E[s^X]=∑k P(X=k)s^k
Recover moments at s=1:
E[X]=φ'(1)
Var(X)=φ''(1)+φ'(1)-(φ'(1))²
Sum of iid: PGF is the product; used in branching.
MGF and random sums
M_X(t)=E[e^{tX}], E[X^k]=M_X^(k)(0)
M_{aX+b}(t)=e^{bt}M_X(at)
For independent X,Y, M_{X+Y}=M_XM_Y. For W=∑_{i=1}^N X_i,
M_W(t)=M_N(log M_X(t)).
E[W]=E[N]E[X],
Var(W)=E[N]Var(X)+Var(N)(E[X])²
N is independent of iid summands X_i; W=0 when N=0. PGF
moments are at s=1; MGF moments are at t=0.

5. MARKOV CHAINS: SETUP

Markov property (memoryless): future depends only on
present state.
P(X_{n+1}=j | X_n=i, past)=P(X_{n+1}=j|X_n=i)
Show a process is Markov
P(X_{n+1}=j|F_n)=P(X_{n+1}=j|X_n)
• Specify the state and possible values; condition on a history
ending at the same current state i.
• Derive the full next-state law and show it depends only on i
(and possibly n). Fresh randomness must be independent of
the history.
To disprove Markov, find two positive-probability histories with
the same present value but different next-event probabilities.
A function of a Markov chain need not be Markov; augment
the state with the needed memory.
State, filtration, specification
S: state space (list of possible states)
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π₀: initial distribution, row vector, entries sum to 1
P: one-step transition matrix, P_{ij}=P(X_{n+1}=j|X_n=i)
Each row of P is a distribution: P_{ij}≥0, rows sum to 1
(stochastic).
F_n: information/history available through time n
X_{n+1}=g(X_n,ζ_{n+1})
If ζ is fresh and independent of the history, then p_{ij}=sum over
z with g(i,z)=j of P(ζ=z). Enumerate inputs, merge equal
destinations, and check boundary rows.
n-step transitions
P⁽ⁿ⁾_{ij}=P(X_n=j|X₀=i)=(Pⁿ)_{ij}
Chapman-Kolmogorov
p^(m+n)_{ij}=∑k p^(m)_{ik}p⁽ⁿ⁾_{kj}
p^(m+n)=p^(m)p⁽ⁿ⁾
Distribution at time n
π_n=π₀ Pⁿ
P(X_n=j)=∑i π₀(i)(Pⁿ)_{ij}
π_b=π_a P^{a,b}, P^{a,b}=P^{a,a+1}...P^{b-1,b}
For time-dependent rules multiply matrices chronologically;
do not replace them by one fixed power.

6. PATHS & INFERENCE

Joint path probability
Multiply start prob by successive one-step probs:
P(X₀=i₀,...,X_n=i_n)=π₀(i₀) P_{i₀ i₁ ... P_{i_{n-1} i_n}}
P(X_{t-0}=i₀,...,X_{t-1}=i_r)=π_{t-0}(i₀)∏i
p_{i_{t-1} i_t}(t-1-t-{1-1})}
Use one-step factors for consecutive times and matrix powers
for gaps; sum over unspecified states.
Bridge / intermediate state
P(X_t=k|X_a=i,X_b=j)=p_{ik}^(t-a)p_{kj}^(b-t)/p_{ij}^(b-a)
For a<t<b and positive denominator, weight the possible
middle state by evidence from both endpoints; the answers
must sum to 1.
Inferring the past (Bayes)
To find P(X₀=i | X_n=j) use Bayes with forward probs:
P(X₀=i|X_n=j)=π₀(i)(Pⁿ)_{ij} / ∑k π₀(k)(Pⁿ)_{kj}
Numerator: prob of that start AND the observation.
Denominator: total prob of the observation.
Time to hit / general recipe
• Condition on the first step (see first-step analysis).
• For 'ever reach' or 'reach before': set up linear equations in
the unknown probabilities.
• Absorbing states give boundary conditions.

7. HIDDEN MARKOV (FILTER)

Hidden chain X_n (matrix P), observations Y_n with emission
P(Y|X)=e_y(y). Track belief π_n(i)=P(X_n=i | Y_{1..n}).
Predict (time update)
pred(j)=∑i π_{n-1}(i)P_{ij}
Update (observation Y_n=y)
π_n(j)∝ pred(j) e_j(y)
π_n(j)=pred(j)e_j(y)/∑k pred(k)e_k(y)
Then normalize so π_n sums to 1. Repeat predict-update per
observation.
Prediction of next / obs
P(X_{n+1}=j|Y_{1..n})=∑i π_n(i)P_{ij}
P(Y_{n+1}=y|Y_{1..n})=∑j pred_{n+1}(j)e_j(y)
HMM path probability
w(x_{0:m})=π₀(x₀)b_{x-0}(y₀)∏_{r=1}^m p_{x_{r-1}x_r}b_{x_r}(y_r)
To smooth a hidden past, sum w over paths with the
requested hidden state and divide by the sum over all paths.
This is a posterior, not an unnormalised joint probability.
7b. STATIONARY DIST.
A distribution π (row vector) is stationary if it is unchanged by
one step:
π P=π , ∑i π_i=1
Solve the linear system (drop one balance equation, use
normalization). For irreducible aperiodic chains π_n→π as n→∞
regardless of π₀.
2-state chain P=[[1-a,a],[b,1-b]]:
π=(b/(a+b), a/(a+b))
Worked filter step
P=[[.7,.3],[.4,.6]], emit e₀=(.9,.1) e₁=(.2,.8), π₀=(.5,.5). Obs y=0
→ π=(.846,.154).

8. STANDARD MODELS
Random walk Steps $Z_i = +1$ w.p. p, -1 w.p. $q = 1 - p$. $S_n = S_0 + Z_1 + \dots + Z_n$. $E[S_n] = S_0 + n(p - q)$ $\text{Var}(S_n) = n(1 - (p - q)^2) = 4npq$ Position after n steps with R rights: $S_n = S_0 + 2R - n$, $R \sim \text{Bin}(n, p)$. $P(S_n = S_0 + 2r - n) = C(n, r) p^r q^{n-r}$ Inventory (s,S) model Restock to S when stock $\leq s$; demand D_n iid. $X_{n+1} = \max(X_n - D_{n+1}, 0)$ then reorder rule Build P from demand distribution; state = stock level. Queue model X_n = customers in system. Arrivals A_n iid, one served per step: $X_{n+1} = \max(X_n - 1, 0) + A_{n+1}$ For capacity N with independent arrival probability λ and service probability μ: interior $p_{i,i+1} = \lambda(1 - \mu)$, $p_{i,i-1} = (1 - \lambda)\mu$, $p_i = \lambda\mu + (1 - \lambda)(1 - \mu)$; boundaries $p_{0,1} = \lambda$ and $p_{N,N-1} = \mu$. Colour-flipping urn $p_{i,i-1} = i/N$, $p_{i,i+1} = (N - i)/N$ i is the current red-ball count in a fixed population N; boundaries force $p_{0,1} = p_{N,N-1} = 1$. General / multidimensional walk $E[S_n] = S_0 + n\mu$, $\text{Var}(S_n) = n\sigma^2$ For a simple symmetric walk in d dimensions, each coordinate moves by ± 1 with probability $1/(2d)$, stays put with probability $1 - 1/d$: $E[x_{n+1}] = 0$ and $\text{Var}(x_{n+1}) = n/d$.
8b. STATE CLASSIFICATION
accessible: j reachable from i: $(P^n)_{ij} > 0$ some n communicate: i, j reach each other; classes partition S irreducible: one class (all states communicate) recurrent: return w.p. 1; else transient period d: gcd of return times; $d = 1$ aperiodic absorbing: $P_{ii} = 1$ (never leaves state i)

9. FIRST-STEP ANALYSIS
Absorbing states a (target) and set. Condition on the first move, then solve linear equations. Hitting probability u_i $u_i = P(\text{hit target } A \text{ before } B \mid \text{start } i)$. Boundary: $u = 1$ on A, $u = 0$ on B. $u_i = \sum_j P_{ij} u_j$ (i transient) Expected hitting time v_i $v_i = E[\text{steps to absorption} \mid \text{start } i]$. Boundary: $v = 0$ on absorbing states. $v_i = 1 + \sum_j P_{ij} v_j$ (i transient) The $+1$ counts the current step. Solve the linear system for all transient i. Expected absorption reward $w_i = r_i + \sum_j P_{ij} w_j$

10. GAMBLER'S RUIN
On $0..N$, absorbing at 0 and N. Up w.p. p, down w.p. $q = 1 - p$. $u_i = P(\text{reach } N \text{ before } 0 \mid \text{start } i)$. Case $p \neq q$ (let $r = q/p$) $u_i = (1 - r^i)/(1 - r^N)$ Case $p = q = 1/2$ (fair) $u_i = i/N$ Prob of ruin (hit 0 first) $= 1 - u_i$ Expected duration v_i Fair game $p = q = 1/2$: $v_i = i(N - i)$ Biased $p \neq q$: $v_i = i/(q - p) - (N/(q - p))(1 - r^i)/(1 - r^N)$ Worked: $p = .4$, $N = 5$ $r = q/p = 1.5$. Reach N before 0: $u_1 \approx .076$, $u_3 \approx .360$. Ruin $= 1 - u_i$.

11. MARTINGALES
$\{M_n\}$ is a martingale if $E[M_{n+1}] = M_n$ and: $E[M_{n+1} M_0..M_n] = M_n$ Then $E[M_n] = E[M_0]$ for all n. Common constructions RW with $p = 1/2$: $M_n = S_n$ is a martingale. $M_n = S_n^2 - n$ (fair RW) is a martingale Biased RW: $(a/n)^{S_n}$ is a martingale. 12. STOPPING TIMES / OST T is a stopping time if $\{T = n\}$ depends only on $X_0..X_n$ (no peeking ahead). Optional Stopping Theorem $E[M_T] = E[M_0]$ Holds if ANY one condition holds: • T is bounded ($T \leq c$), or • M is bounded and $T < \infty$ a.s., or • $E[T] < \infty$ and increments bounded. Use: apply $E[M_T] = E[M_0]$ to solve exit probs / expected times.

13. BRANCHING (GALTON-WATSON)
Z_n = size of generation n, $Z_0 = 1$. Each individual has iid offspring, PGF ϕ. X: offspring count per individual m: mean offspring $E[X] = \phi'(1)$ σ^2: offspring variance $\text{Var}(X)$ Generation PGF (compose) $\phi_{n+1}(s) = \phi_n(\phi(s)) = \phi(\phi_n(s))$ Moments $E[Z_n] = m^n$ Variance ($m \neq 1$): $\text{Var}(Z_n) = \sigma^2 m^{n-1}(m^n - 1)/(m - 1)$ If $m = 1$: $\text{Var}(Z_n) = n\sigma^2$. Extinction prob q q = smallest root in $[0, 1]$ of: $q = \phi(q)$ • $m \leq 1 \Rightarrow q = 1$ (certain extinction). • $m > 1 \Rightarrow q < 1$ (survival possible). Vaccination / thinning Vaccinate fraction v: effective mean $m' = m(1 - v)$. Outbreak dies if $m' \leq 1$. $v^* = 1 - 1/m$ (herd-immunity threshold) Worked: Poisson(m) offspring $\phi(s) = e^{m(s-1)}$. $m = 1.5$ gives extinction $q = 0.417$, herd $v^* = 1/3$.